

**Differential Pair Layout
using
TSMC 65nm Technology**

Differential Pair Schematic

AGND

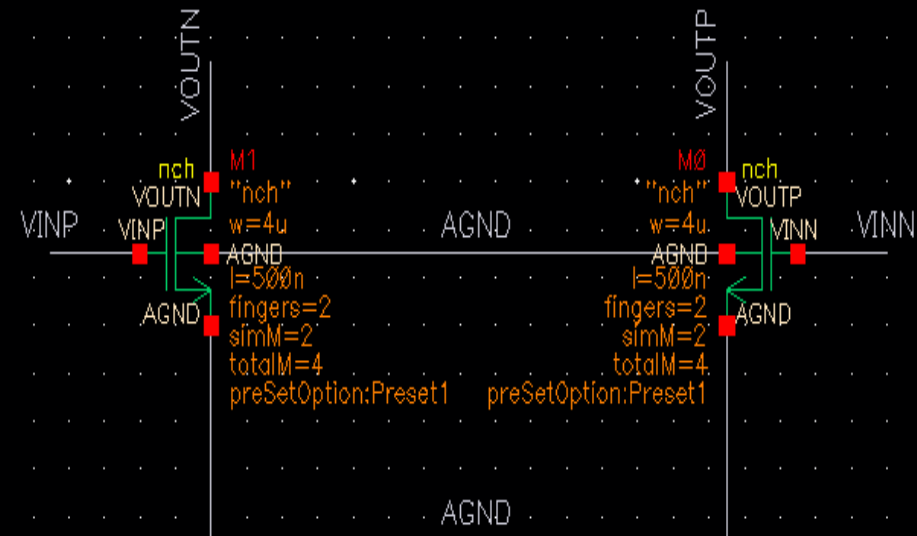
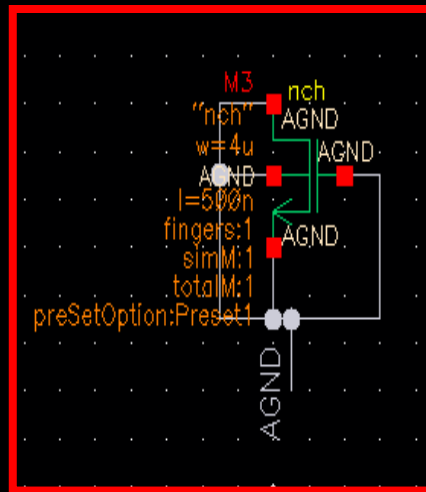
VINN

VINP

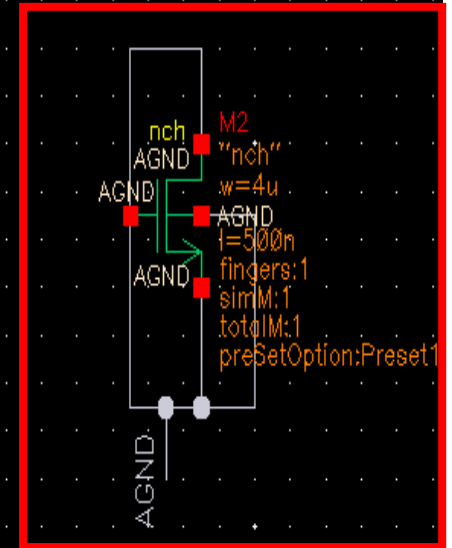
VOUTP

VOUTN

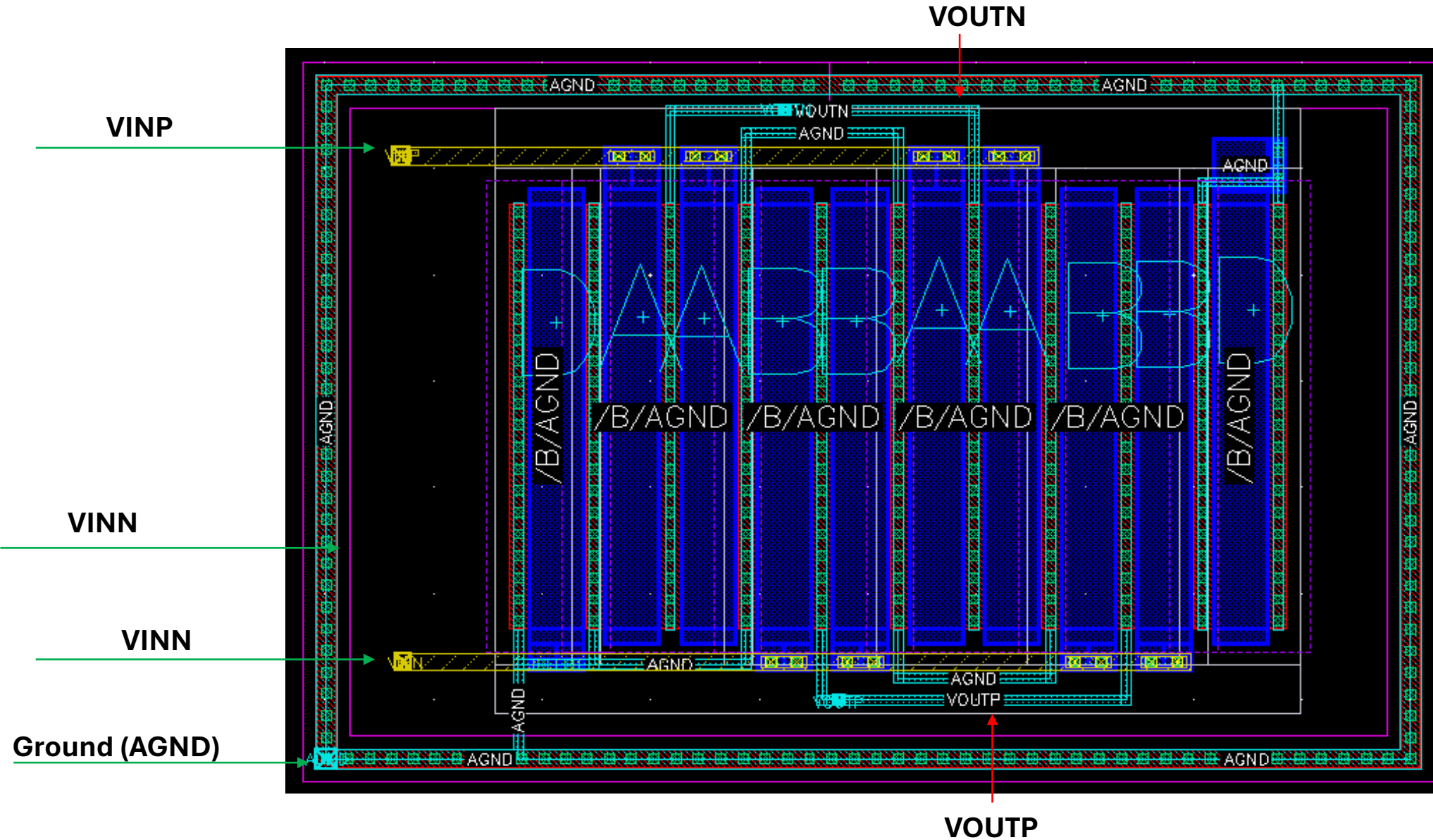
Dummy Device



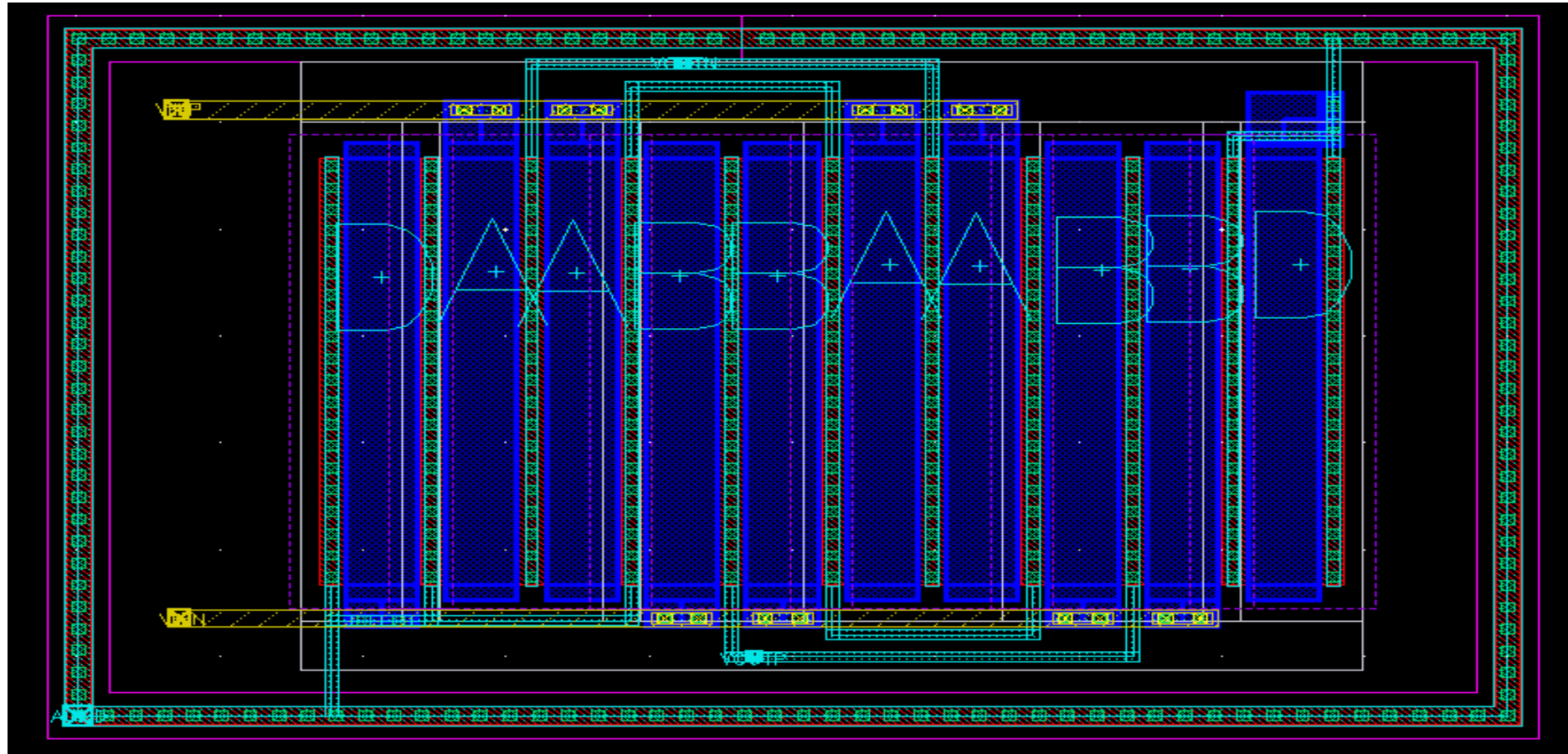
Dummy Device



Differential Pair Layout



Differential Pair Layout Floor Plan



D

A

A

B

B

A

A

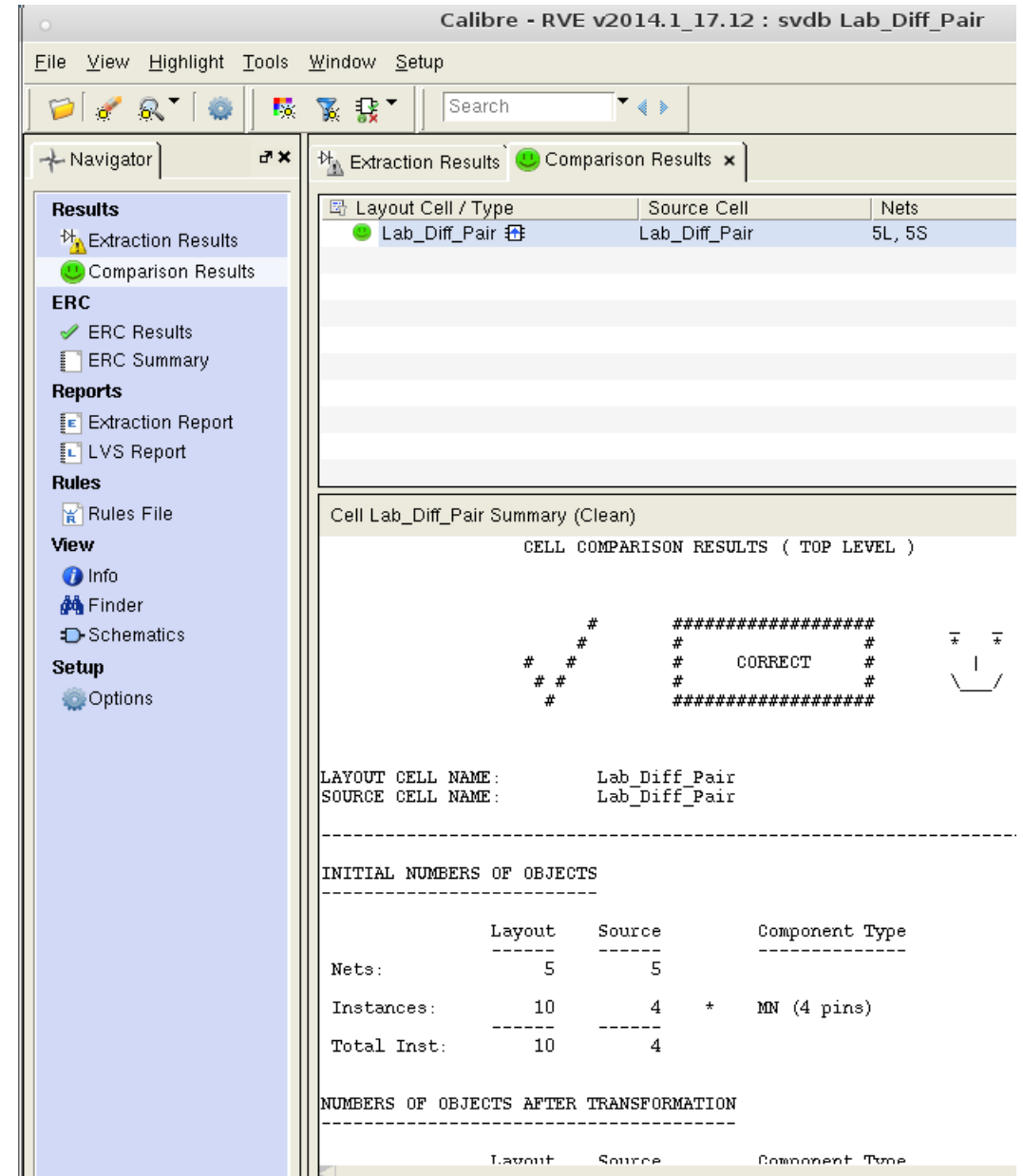
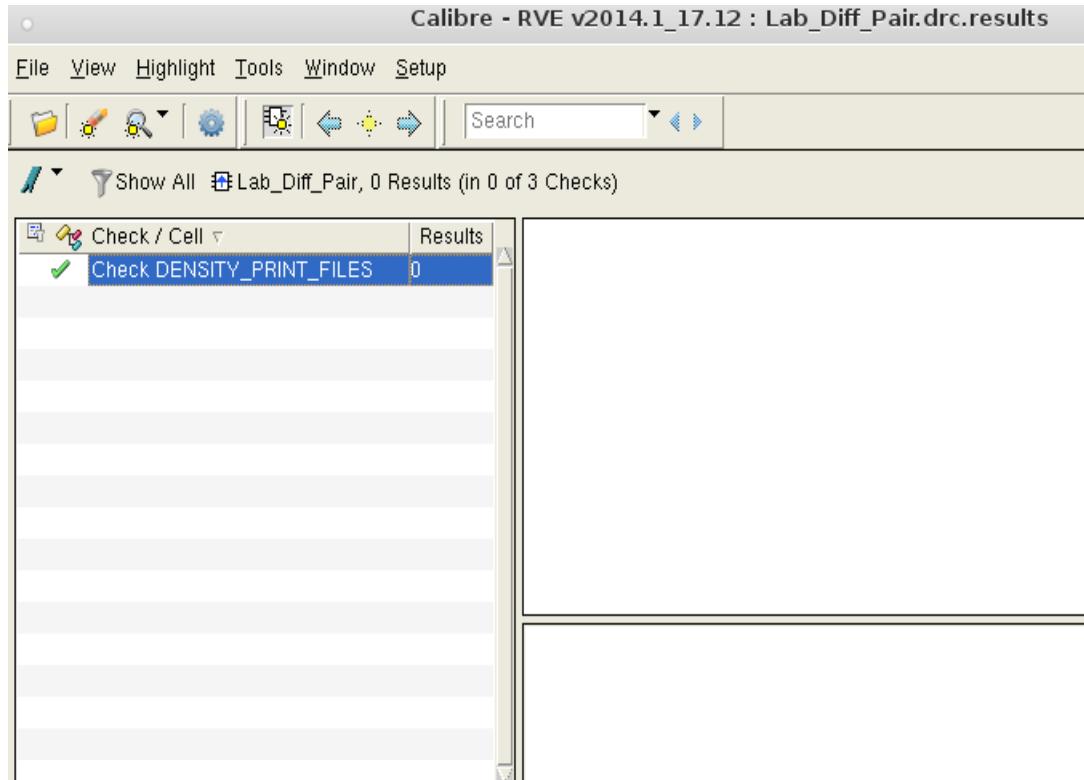
B

B

D

- The Layout technique for differential pair is interdigitation .
- This will give matching for the routing .

DRC and LVS



PEX and Result Viewing Environment (RVE)

Calibre Interactive - PEX v2014.1_17.12

File Transcript Setup

Rules
Inputs
Outputs
PEX Options
Run Control
Transcript
Run PEX
Start RVE

```
--- NETWORK REDUCTION BEGIN:
--- READING FROM PDB...
--- BEGIN REDUCING NETS...
--- DONE REDUCING NETS...
--- WRITING TO PDB...
--- NETWORK REDUCTION COMPLETE: CPU TIME = 0 REAL TIME = 0 LVHEAP = 329/331/331 MALLOC = 340/340/340

-----
PDB NET SUMMARY
-----
pdb file name = svdb/LAB_DIFF_PAIR.pdb
root cell name = LAB_DIFF_PAIR
total nets = 5
top-level nets = 5
non-top-level nets = 0
degenerate nets = 0
merged nets = 0
error nets = 0

-----
CALIBRE xRC WARNING / ERROR Summary
-----
xRC Warnings = 1
xRC Errors = 0
-----

--- CALIBRE xRC::FORMATTER COMPLETED - Thu Jul 30 13:56:11 2026
--- TOTAL CPU TIME = 0 REAL TIME = 0 LVHEAP = 26/55/331 MALLOC = 307/307/340 ELAPSED
```

Calibre Info

Calibre View generation completed with 0 WARNINGS and 0 ERRORS. Please consult the CIW transcript for messages.

Close

Calibre - RVE v2014.1_17.12 : svdb Lab_Diff_Pair

File View Highlight Tools Window Setup Help

Navigator

Lab_Diff_Pair

Results

- Extraction Results
- Comparison Results
- Parasitics

No.	Layout Net	Source Net	R Count	C Total (F)	CC Total (F)	C+CC Total (F)
1	VINP	VINP	16	4.13069E-16	4.00799E-15	4.42106E-15
2	VINN	VINN	16	4.12263E-16	4.11686E-15	4.52912E-15
3	AGND	AGND	188	2.46553E-15	5.01036E-15	7.47589E-15
4	VOUTN	VOUTN	36	1.32851E-16	2.55927E-15	2.69212E-15
5	VOUTP	VOUTP	36	1.41824E-16	2.37896E-15	2.52079E-15